



SWARM

Dock v0 · First Autonomous Node Infrastructure

**The dock is not the product.
The dock is the node.**

01 · Core Principle

The first SWARM dock is not futuristic infrastructure.

The objective is:

- reliability
- low cost
- autonomous deployment
- weather protection
- charging
- telemetry relay

Minimum viable infrastructure.

02 · Strategic Direction

SWARM does not initially build proprietary drones.

Initial stack:

- DJI-compatible drones
- SDK integration
- SWARM OS coordination layer
- semi-custom dock

The moat remains software and orchestration.

03 · First Dock Architecture

Initial dock structure:

- industrial weather-resistant enclosure
- automated lid or hatch
- charging pad or contact charging
- local compute node
- telemetry relay
- cooling and power management

Drones remain docked until event-triggered deployment occurs.

04 · What The Dock Must Do

Required capabilities:

- protect drone from environment
- maintain charging state
- enable autonomous launch
- enable autonomous return
- maintain network connection
- relay telemetry to SWARM OS

Nothing more initially.

05 · What NOT To Build Initially

Avoid:

- robotic battery swap
- complex robotic arms
- multi-drone launch systems
- sci-fi mechanics
- expensive industrial automation

Complexity kills early-stage robotics startups.

06 · Environmental Constraints

The real challenge is operational reliability.

The dock must survive:

- rain
- wind
- dust
- heat
- humidity
- unstable connectivity

Real-world reliability matters more than futuristic aesthetics.

07 · Example Lean Deployment

Example deployment:

- 3 autonomous micro-units
- 1 SWARM Dock v0
- environmental sensors
- local compute node

Estimated initial deployment: €5k–15k.

08 · Long-Term Evolution

Future dock evolution:

- edge AI compute
- autonomous battery swap
- multi-unit coordination
- mesh networking
- distributed territorial coverage

The dock evolves from enclosure into infrastructure node.

